Monotone Strategy-Specific Erosion under Volatility-Targeting Crowding: Matched-Control Identification via Agent-Based Simulation*

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Abstract

Volatility targeting (VT), trend-following (TF), and short-horizon mean-reversion (MR) share a procyclical structure: each strategy's position update is a non-trivial function of recent prices or volatility, so widespread adoption can in principle generate self-reinforcing trading flows. Whether the resulting strategy-performance degradation reflects a discrete crowding tipping point, and whether it is VT-specific or attributable to crowded flow in general, remains unsettled. We build an agent-based model with 1,000 agents of heterogeneous types, a Kyle (1985) market maker, endogenous VIX dynamics, and 200 fixed noise traders, and re-examine the threshold structure under matched microstructure with an exogenous sup-Wald detector and turnover-matched random-direction controls. Across 94,500 Monte Carlo simulations in the exogenous-detector redesign layer (27 cell–adoption combinations—seven adoption levels in the baseline cell and five in each of four perturbed cells—$\times$ 7 treatments/controls $\times$ 500 MC), three findings emerge. First, VT Sharpe declines *monotonically* with adoption rather than crossing a discrete threshold: in the canonical cell ($\lambda = 0.005$, $\gamma = 200$), Sharpe falls from 0.510 at 10% adoption to -0.271 at 100%, with adjacent path-bootstrap 95% confidence intervals separating from 40%

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onward; the sup-Wald test rejects flatness in all five cells ($p = 0.001$ in every cell) but identifies no interior break (the single breakpoint localizes to the saturation boundary), and the maximum marginal degradation concentrates in $(70\%, 100\%]$. Second, a turnover-matched random-direction control (RR_VT) matching VT's trading-footprint distribution without its vol-feedback direction exhibits no degradation in any cell; at VT's footprint scale, the erosion is therefore identified to the *systematic direction* of vol-feedback trading rather than to crowded flow, providing a cleaner mechanism counterfactual than a no-strategy baseline (at trend-following footprint scales two orders of magnitude larger, direction-randomized controls do erode, so the identification is footprint-scale-dependent). Third, the canonical "70% threshold" reported under a Sharpe-only detector survives only as a descriptive relative-drop level-crossing (drop $> 70\%$) in 3 of 5 cells; under the exogenous, treatment-aware detector the interpretation is monotone erosion without a tipping point. The applicability gate excludes the TF treatment in four of five cells and MR in all five (structurally loss-making regimes), so cross-strategy family claims are withdrawn and the paper's mechanism evidence is confined to VT; our results sharpen the policy-relevant question from "where is the tipping point" to "how steep is the monotone erosion and what identifies its mechanism."

Keywords: volatility targeting, positive feedback, crowding, agent-based model, matched-control identification, monotone erosion

JEL Classification: G11, G12, G17, G18

1 Introduction

Volatility targeting (VT) has become a ubiquitous tool in dynamic portfolio management. The canonical framework, formalized by Moreira and Muir [2017], prescribes scaling equity exposure as $w_t = \sigma^*/\hat{\sigma}_t$, where σ^* is the target volatility and $\hat{\sigma}_t$ is a conditional volatility estimate. Harvey et al. [2018] demonstrate that such strategies meaningfully reduce tail risk across asset classes, and the approach has been widely adopted by pension funds, insurance companies, and target-date funds. Industry estimates suggest that short-volatility and volatility-sensitive strategies collectively manage over USD 2 trillion in assets [Cole, 2017]. VT is, however, only the largest individual class within a wider family of positive-feedback strategies—including trend-following [Moskowitz et al., 2012, Asness et al., 2013] and short-horizon mean-reversion [Lehmann, 1990, Lo and MacKinlay, 1990]—that share a procyclical risk-of-volatility structure and could in principle exhibit similar crowding behavior.

Despite individual-level benefits, all three strategy classes embed a fundamentally procyclical mechanism: when volatility rises (or trends turn, or signals saturate), all adopters simultaneously rebalance in the same direction, creating correlated trading pressure. The crowding risk literature warns of such procyclical dynamics—Baltas [2019] documents crowding effects in alternative risk premia, the European Central Bank [ECB, European Central Bank, 2020] explicitly flagged VT procyclicality as a source of market fragility, and Cont and Bouchaud [2000] build an agent-based herd-behavior model in which correlated demand generates fat-tailed returns. The closely related fire-sale literature [Greenwood and Thesmar, 2011] shows that forced selling by one investor class concentrates price impact on overlapping holdings and can amplify subsequent declines, a mechanism we view as the continuous-time counterpart of the adoption-axis erosion we estimate. None of these studies, however, provides a *strategy-class-by-strategy-class* quantitative threshold. Several commentators draw parallels to the portfolio insurance strategies implicated in the 1987 crash [Gennotte and Leland, 1990]. Bookstaber et al. [2014] use agent-based modeling to study financial system fragility broadly, while LeBaron [2006] and Danielsson et al. [2012] study feedback-driven market dynamics; applications that *ex-*

plicitly compare VT crowding against TF or MR crowding under identical microstructure are absent.

A critical gap remains: *does correlated positive-feedback trading produce a discrete crowding tipping point, and—if degradation is observed—is it attributable to the systematic trading direction of the strategy or to crowded flow in general?* Existing analyses are either qualitative [European Central Bank, 2020], focused on a single strategy without a counterfactual separating trading footprint from trading direction [Baltas, 2019], or lack a quantitative threshold altogether. The empirical VT-alpha literature itself is contested—Cederburg et al. [2020] and Liu et al. [2019] question whether VT's Sharpe improvement survives realistic implementation costs, while Barroso and Detzel [2021] find that volatility-managed market portfolios survive transaction costs in directions opposite to those questioned by Cederburg et al. [2020]—but this debate is orthogonal to our question: we study the crowding externality *conditional* on a strategy class being used, not whether any individual strategy delivers alpha. The mechanism-identification question we pose—separating crowded flow from systematic direction—requires a control that matches the offending strategy's trading footprint without inheriting its directional rule, a comparator absent from the prior literature.

We address this gap with an agent-based simulation in which the same Kyle-microstructure (constant λ, endogenous VIX) is exposed to a panel of strategy treatments (VT, TF, MR), three turnover-matched random-direction controls (RR_VT, RR_TF, RR_MR) that match each treatment's trading-footprint distribution—rebalance frequency and the moments of $|\Delta w|$, measured per cell–adoption pair—while randomizing its direction, and a NoiseControl anchor verifying detector specificity. Detector inference is exogenous: a sup-Wald test with a flatness-calibrated permutation null over the adoption axis, plus path-bootstrap confidence intervals at each adoption level (M = 500 simulations per cell; §3.5). Our contribution is threefold. First, we establish *monotone strategy-specific erosion* as the empirical signature of VT crowding rather than a discrete tipping point: in the canonical cell ($\lambda = 0.005$, $\gamma = 200$), VT Sharpe declines from 0.510 at 10% adoption to -0.271 at 100%, the sup-Wald test rejects flatness in all five microstructure cells ($p = 0.001$

in every cell), adjacent path-bootstrap 95% CIs separate from 40% onward, and the maximum marginal degradation localizes to $(70\%, 100\%]$; the canonical "70% threshold" survives only as a descriptive relative-drop level-crossing in 3 of 5 cells. Second, we identify the erosion mechanism to the *systematic direction* of vol-feedback trading at VT's footprint scale: the turnover-matched random-direction control RR_VT, which inherits VT's footprint distribution without its directional rule, exhibits zero degradation in any cell (Sharpe in fact improves by +0.080 to +0.098 with adoption), isolating the externality to vol-feedback signal direction rather than to crowded flow at that scale. This control delivers a cleaner mechanism counterfactual than the no-strategy NoiseControl baseline of prior work; the scale qualifier matters, because at trend-following footprint scales the direction-randomized controls themselves erode significantly (§3.5). Third, we report *joint robustness* of the qualitative monotone pattern across the five microstructure cells, which span ±50% perturbations of Kyle-λ and VIX-feedback γ around the canonical baseline. Matched-control evidence on TF and MR is constrained by extensive matched-input gate failures—the TF treatment is structurally loss-making in four of five cells and MR in all five at $s = 10$—so family-level ordering claims are withdrawn and interpreted accordingly in §3.5. The fixed-vs-scaled-liquidity validation that featured in earlier work is retained as a methodological check (§3.8) and confirms that approximately half of the originally observed degradation is attributable to liquidity evaporation rather than to crowding per se. These results sharpen the policy-relevant question from "at what adoption is the tipping point" to "how steep is the monotone erosion, and what identifies its mechanism."

2 Model

We construct an agent-based model with $N = 1{,}000$ agents of heterogeneous types operating in a single risky asset over $T = 2{,}520$ trading days (approximately 10 years).

2.1 Agent Types and Strategy Treatments

The model supports four interchangeable *strategy treatments* operating under identical microstructure (Eqs. 4–5). For a given experimental run, the population is partitioned as $N_{\text{BH}} = (1-\phi)N - N_{\text{noise}}$ Buy-and-Hold (BH) agents with constant weight $w^{\text{BH}} = 1.0$, ϕN strategy-treatment agents, and $N_{\text{noise}} = 200$ noise traders with weight increments $\Delta w \sim N(0, 0.02)$ clipped to $[0, 1.5]$ providing background liquidity. The composition is well-defined as long as $\phi < 1 - N_{\text{noise}}/N = 80\%$; at the saturated adoption levels $\phi \in \{100\%\}$ all non-noise agents are strategy-treatment agents.

The strategy treatment is one of four classes:

Volatility targeting (VT). The canonical 12/VIX rule [Perchet et al., 2015],

$$w_t^{\text{VT}} = \min\bigl(12/\text{VIX}_{t-1},\ 1.5\bigr), \tag{1}$$

with the lagged VIX ensuring lookahead-safe execution.

Trend-following (TF). Momentum-based weight update,

$$w_t^{\text{TF}} = \text{clip}\bigl(0.5 + s \cdot \tilde{r}_{t-1}^{(W)},\ 0,\ 1.5\bigr), \tag{2}$$

where $\tilde{r}_{t-1}^{(W)}$ is the W-day cumulative log-return up to $t-1$ (lookahead-safe), s is a scaling intensity, and the clip enforces the same $[0, 1.5]$ rail as VT. Baseline values $s = 10$ and $W = 22$ correspond to one-month time-series momentum at intermediate scaling consistent with cross-asset CTA practice [Moskowitz et al., 2012]; the strategy-spec robustness analysis (§3.5) sweeps $s \in \{1, 3, 5, 10\}$ and $W \in \{10, 22, 60\}$.

Mean-reversion (MR). Sign-flipped momentum,

$$w_t^{\text{MR}} = \text{clip}\bigl(0.5 - s \cdot \tilde{r}_{t-1}^{(W)},\ 0,\ 1.5\bigr), \tag{3}$$

with the same baseline $s = 10$, $W = 22$. The negative coefficient captures the short-horizon reversal documented by Lehmann [1990] and Lo and MacKinlay [1990]. In our agent-based environment, MR's positive-feedback channel arises asymmetrically: when prices fall, the strategy buys, but if a sufficiently large fraction of agents trade in the same direction, the buying pressure itself can push prices high enough to trigger a reverse-direction signal cascade. This produces episodic price-collapse-and-recovery behavior at intermediate adoption (e.g. final prices on the order of 10^{-23} at $\phi = 30\%$ in 500/500 simulations from the Phase 1 run — a legitimate simulation finding rather than a numerical artifact, documented in the threshold caveats of §3.5).

NoiseControl (NC). Falsifiability anchor with deterministic weight $w_t^{\mathrm{NC}} = 0.5$. The 0.5 weight matches the noise-trader baseline (mean weight 0.5 from $\Delta w \sim N(0, 0.02)$ random walk anchored at 0.5), ensuring NoiseControl's mean order-flow contribution matches that of the noise traders rather than that of BH agents (whose constant unit weight would inflate aggregate order flow under high ϕ). This is the strictest falsifier choice: if even matching-noise behavior produced a threshold, our positive-feedback claim would be undermined. The NoiseControl treatment replaces VT/TF/MR agents with constant-weight agents whose order flow is statistically indistinguishable from the noise-trader background. Under the null hypothesis that crowding is a generic mechanical artifact of large coordinated trading—rather than a positive-feedback property—NoiseControl should still produce a critical adoption threshold; under our maintained hypothesis it should not. The Phase 1 run (§3.5) confirms NoiseControl produces no detectable threshold across $\phi \in \{0\%, \ldots, 100\%\}$, establishing that the cross-treatment threshold differences reflect the strategies' positive-feedback structure rather than their headcount.

Random-direction matched controls (RR_VT, RR_TF, RR_MR). The NoiseControl anchor establishes that headcount alone does not produce crowding, but it cannot identify whether the residual treatment-specific erosion is driven by (i) the systematic *direction* of vol-feedback / momentum / mean-reversion signals, or (ii) the sheer correlated *footprint* of high-turnover trading regardless of direction. To separate these two channels, we introduce a stronger counterfactual that holds the treatment's trading-footprint *distribution* fixed but randomizes its directional rule. For each treatment $X \in \{\text{VT, TF, MR}\}$, the matched-control RR_X at adoption ϕ inherits the realized trading-footprint *distribution* of the corresponding X treatment—its rebalance frequency and the mean and standard deviation of its per-step weight-change magnitude $|\Delta w^X|$, measured from the treatment runs in the same (cell, adoption) pair—then applies the sign of each Δw_t from an independent Bernoulli(0.5) draw, retaining the original clip rail $[0, 1.5]$. Under the maintained hypothesis that crowding is mechanism-driven (a function of systematic positive-feedback direction), RR_X should exhibit no degradation despite matching X's footprint; under the null that any large coordinated turnover destabilizes the market, RR_X should reproduce X's degradation. RR controls deliver a cleaner mechanism counterfactual than NoiseControl because the comparator is endogenous to the treatment rather than a fixed deterministic baseline.

Matched-input applicability gate. An RR_X control is informative only insofar as the treatment X from which it inherits its footprint is operating in a non-pathological regime: if X collapses the market (e.g. MR at intermediate adoption can drive end-of-horizon prices toward zero in a fraction of simulations), the realized $|\Delta w_t^X|$ becomes dominated by collapse-driven extremes whose post-clip distribution no longer matches the treatment's design intent, and RR_X inherits this distorted footprint rather than a representative one. We therefore introduce a *matched-input gate*: an RR_X result in cell c at adoption ϕ is reported as applicable only when the treatment X at the same (c, ϕ) passes a baseline-Sharpe floor: mean Sharpe across MC simulations > -0.5, ruling out structurally loss-making regimes where the treatment itself is not viable. Cells failing this condition are excluded from the matched-control conclusions and noted as *matched-control not applicable*; the standalone treatment evidence in those cells is retained but interpreted with explicit pathological-regime caveats (§3.5). The gate is implemented in the replication package and applied uniformly across all RR_VT/RR_TF/RR_MR comparisons.

The experimental adoption variable ϕ now denotes the fraction of N agents using the *active* strategy treatment, tested at seven levels: 0%, 10%, 20%, 30%, 50%, 70%, and 100%. As ϕ rises, BH agents are replaced by treatment agents while noise traders remain fixed at $N_{\text{noise}} = 200$. This design isolates the crowding mechanism from liquidity effects: because the noise-trader population is constant across all adoption levels, any performance degradation is attributable to correlated strategy-treatment trading rather than to the mechanical reduction of background liquidity providers. Section 3.8 reports a design validation confirming that this distinction is quantitatively important. Cross-treatment seed pairing is enforced—seed$(i) = \lfloor \phi \cdot 10^5 \rfloor + i + 42$—so that Monte-Carlo sampling noise is paired across VT, TF, and MR comparisons within each adoption level.

2.2 Market Microstructure

Price formation follows a simplified Kyle [1985] model:

$$r_t = \mu + \lambda \cdot \frac{\text{NOF}_t}{N} + \varepsilon_t, \quad \varepsilon_t \sim N(0, \sigma_f), \tag{4}$$

where $\mu = 0.08/252$ is the daily drift, $\lambda = 0.005$ is the Kyle lambda (price impact per unit of normalized order flow), NOF_t is the aggregate net order flow from all agents, and $\sigma_f = 0.16/\sqrt{252}$ is the fundamental volatility. The VIX evolves endogenously:

$$\text{VIX}_t = \max\Big(0,\ \text{VIX}_{t-1} + \kappa\big(\bar{V} + \gamma \cdot \Delta\sigma_t^{\text{real}} - \text{VIX}_{t-1}\big) + \eta_t\Big), \tag{5}$$

where $\kappa = 0.03$ is the mean-reversion speed, $\bar{V} = 18$ is the long-run mean, $\Delta\sigma_t^{\text{real}} = \max(0,\ \hat{\sigma}_t^{20d} - 0.16)$ is the positive part of the deviation of 20-day realized volatility from the baseline annualized volatility of 16%, $\gamma = 200$ amplifies this deviation, and $\eta_t \sim N(0, 0.3)$. The asymmetric $\max(0, \cdot)$ ensures that VIX responds to volatility increases but not decreases, capturing the empirical observation that VIX rises faster than it falls. The VIX is additionally bounded to $[9, 80]$.

2.3 Feedback Structure

The model is designed to embed a positive feedback loop: when VT agents sell (because VIX rises), the negative order flow depresses prices via Eq. (4), which increases realized volatility, which raises VIX via Eq. (5), which triggers further VT selling. This circular structure is a deliberate implementation of the Brunnermeier and Pedersen [2009] loss spiral applied to the VT context; it is a modeling assumption, not a simulation output. The strength of this feedback loop is proportional to ϕ, and the simulation's purpose is to quantify how steeply the loop's performance consequences accumulate as adoption rises—not to locate an adoption level at which it switches on.

2.4 Simulation Design

We organize the simulation evidence base into three layered phases. **Phase 1** (cross-treatment baseline at cell1 microstructure $\lambda = 0.005$, $\gamma = 200$) runs $M = 500$ independent Monte Carlo simulations across all four treatments (VT/TF/MR/NoiseControl) at the 7 adoption levels enumerated above—14,000 simulations in total, of which the 3,500-simulation VT slice reuses the earlier VT baseline run under identical settings (the other 10,500 are new) and supplies the standalone VT analysis (Tables 2–4). Bootstrap 95% confidence intervals (2,000 replications) are reported for all key VT metrics. **Phase 2** (strategy-spec robustness) extends the comparison to TF and MR under a 12-cell scaling-by-window grid ($s \in \{1, 3, 5, 10\}$, $W \in \{10, 22, 60\}$) at 7 adoption levels with $M = 100$ MC per cell, totaling 16,800 simulations (Table 7). **Phase 2b** (microstructure OAT) crosses the four treatments under five (λ, γ) cells (cell1 baseline plus λ/γ each at $\pm50\%$ of baseline; κ is held fixed because this run, designed after Phase 1's then-current cross-treatment comparison, prioritizes the impact-and-feedback parameters most relevant to the Kyle mechanism) at four adoption levels ($\phi \in \{10\%, 30\%, 70\%, 100\%\}$) with $M = 200$ MC per (treatment, ϕ) combination, totaling 16,000 simulations distributed across the 5 OAT cells (Table 8). The three phases combine to 46,800 simulations spanning approximately 118 million agent-day observations. Finally, the **exogenous-detector redesign layer**—the evidence layer behind every headline claim in Section 3—resimulates all seven treatments and controls (VT, TF, MR, RR_VT, RR_TF, RR_MR, NoiseControl) at $M = 500$ MC per (cell, adoption) point on a denser adoption grid: $\phi \in \{10, 30, 40, 50, 60, 70, 100\}\%$ in the canonical cell and $\phi \in \{10, 30, 50, 70, 100\}\%$ in each of the four perturbed cells, i.e. 27 cell–adoption combinations and $27 \times 7 \times 500 = 94{,}500$ simulations. The random-direction controls are *coherent-block*: on each day a single random draw applies one common sign to the block's entire matched $|\Delta w|$ weight vector, so the control reproduces the treatment's aggregate, coordinated footprint—per-agent independent randomization would diversify away in aggregation and understate the footprint being matched. All simulation seeds are deterministic (a fixed base formula plus treatment-specific offsets that keep treatment and control seed streams disjoint), so every run is exactly reproducible. For each simulation we compute: annualized market volatility, strategy-treatment Sharpe ratio (annualized mean excess return over risk-free rate divided by annualized standard deviation of daily returns), maximum drawdown, excess kurtosis, skewness, flash-crash frequency (daily returns exceeding 3σ), and VIX dynamics.

3 Results

3.1 Strategy Performance Degradation: Monotone, Not Tipping

Table 2 presents the core VT degradation results under fixed background liquidity ($N_{\text{noise}} = 200$). The pattern is best characterized as *monotone, strategy-specific erosion*—a continuous decline in Sharpe with adoption rather than a discrete threshold crossing. Adjacent path-bootstrap 95% confidence intervals separate from $\phi = 40\%$ onward (Table 1), the exogenous sup-Wald flatness test rejects ϕ-invariant Sharpe in all five microstructure cells ($p = 0.001$ in every cell; full per-cell statistics in the replication package), and the maximum marginal degradation localizes to the $(70\%, 100\%]$ interval. The cell-level rejections survive multiple-testing discipline: across the redesign layer's 35 (cell, treatment) detector runs, a Bonferroni correction leaves $p = 0.001$ significant ($35 \times 0.001 = 0.035 < 0.05$), while marginal readings at $p = 0.003$ would not survive and are not relied upon.[1]

In the canonical cell ($\lambda = 0.005$, $\gamma = 200$, $M = 500$ Monte Carlo paths), VT Sharpe falls from 0.510 at 10% adoption to -0.271 at 100% (a -0.78 absolute change spanning the adoption axis), passing through 0.481 at 30%, 0.405 at 40%, 0.338 at 50%, 0.203 at 60%, and 0.091 at 70%. The 30–50% decline that the earlier Sharpe-only narrative attributed to a "safe-zone-to-tipping" transition is, under the exogenous detector with denser adoption grid, the middle portion of an already-significant monotone slope; the 70–100% interval contains the steepest single-step drop ($0.091 \to -0.271$) but the path-bootstrap CIs of all adjacent levels from 40% onward are non-overlapping, so the steepness pattern is one of accelerating monotone erosion rather than knife-edge transition. Table 1 reports the full canonical-cell curve.

[1] An earlier version of this paper, calibrated against a Sharpe-only detector that itself used adoption-level performance to define both the threshold and the falsification criterion, reported a discrete "70% tipping point." The exogenous-detector resimulation ($M = 500$, documented in the replication package) establishes that the sup-Wald rejection of flatness is consistent with smooth monotone decline and does not localize an interior break (the argmax breakpoint sits at the saturation boundary, after 70%): the permutation null tests ϕ-invariance, not the absence of a break, so the resulting $p = 0.001$ in every cell identifies non-flatness but not a discrete threshold. The headline "70%" figure survives only as a descriptive level-crossing (relative Sharpe drop $> 70\%$) in 3 of 5 microstructure cells and is not the output of a structural break detector. The exogenous-detector narrative supersedes the earlier tipping-point framing throughout this paper.

Table 1: Canonical-cell VT Sharpe under adoption-axis monotone erosion (exogenous-detector resimulation, $M = 500$).

ϕ	VT Sharpe (mean)	95% Path-Bootstrap CI	Δ vs. previous level	CI overlaps previous?
10%	0.510	[0.484, 0.534]	—	—
30%	0.481	[0.454, 0.506]	−0.029	yes
40%	0.405	[0.380, 0.427]	−0.076	no
50%	0.338	[0.313, 0.362]	−0.067	no
60%	0.203	[0.182, 0.225]	−0.135	no
70%	0.091	[0.073, 0.108]	−0.112	no
100%	−0.271	[−0.283, −0.260]	−0.362	no

Path-bootstrap CIs use 2,000 resamples within each adoption cell drawn over MC simulation paths (not within-path days), so the reported widths reflect between-simulation Sharpe dispersion. The 10%–30% step is the only adjacent pair whose CIs overlap; from 40% onward, no adjacent CI pair overlaps, providing statistical evidence of continuous monotone decline. The grid spacing 70%–100% is the widest in the adoption axis (30 pp versus 10 pp elsewhere); the −0.362 step there should be read as the maximum marginal degradation *rate per unit adoption* only after dividing by the spacing, which yields a per-10pp rate of −0.121 comparable to the 60%–70% step (−0.112 per 10pp) rather than a discrete jump.

Table 2 provides the legacy results table summarising VT strategy and market outcomes on the original adoption grid (0, 10, 20, 30, 50, 70, 100%); the qualitative pattern there—essentially flat below 30%, sharp drop to negative Sharpe by 100%—remains correct under the new framing, but should be read as a sparse-grid view of the monotone erosion documented in Table 1 rather than as evidence of a discrete tipping point. The narrow Sharpe interval at $\phi = 10$–30% in Table 2 corresponds to the leftmost segment of the monotone curve where decline rate per unit adoption is still small; the sharp drop from 50% to 70% in Table 2 reflects the same monotone slope steepening through the dense-grid 40%, 50%, 60% intermediate levels shown in Table 1.

Table 2: VT Strategy and Market Outcomes by Adoption Level

	Strategy Performance				**Market Stability**			
ϕ	Sharpe	95% CI	ΔSharpe	Ret. (%)	ΔVol (%)	Kurt.	Kurt. CI	Flash/yr
0%	—	—	—	—	baseline	−0.00	[−0.01, 0.01]	0.32
10%	0.47	[0.44, 0.50]	—	4.69	−0.1	−0.01	[−0.02, −0.00]	0.30
20%	0.50	[0.47, 0.52]	+6%	4.98	+0.8	0.00	[−0.01, 0.01]	0.31
30%	0.47	[0.44, 0.50]	−0%	4.74	+3.8	−0.00	[−0.01, 0.00]	0.31
50%	0.34	[0.31, 0.36]	−28%	3.46	+18.8	0.06	[0.05, 0.07]	0.40
70%	0.08	[0.07, 0.10]	−82%	0.90	+42.7	1.41	[1.28, 1.55]	1.09
100%	−0.27	[−0.28, −0.26]	−157%	−3.82	+119.1	61.4	[59.2, 63.4]	1.20[a]

Results are means across 500 Monte Carlo simulations × 2,520 days each (N = 1,000 agents; N_{noise} = 200 fixed). 95% CIs from 2,000 bootstrap replications drawn iid from the pooled return distribution within each adoption cell (∼1.26M-day pool per cell). Monte Carlo standard error across the 500 sim-level Sharpe estimates is ≤ 0.02 for all adoption levels; reported CIs are bootstrap-within-pool, so the Table 2 CI widths reflect within-cell return-distribution dispersion rather than between-simulation Sharpe variability. Sharpe ratio and return are for the VT portfolio. ΔSharpe is relative to the 10% baseline. ΔVol is the percentage change in annualized market volatility relative to the 0% (no VT) scenario. Flash crash frequency counts daily returns > 3σ. Kurtosis is excess (Fisher); bootstrap CIs reported for all rows. The narrow kurtosis CI at $\phi = 100\%$ ([59.2, 63.4] on a 61.4 estimate) reflects the very large pooled-return sample (1.26M days × 500 sims) under iid bootstrap; a block-bootstrap robustness with block length 10 days widens the CI by less than ±1.5 kurtosis units and does not change the qualitative two-orders-of-magnitude jump from 70% to 100%.

[a] At 100% adoption, the moderate flash crash frequency relative to 70% reflects a measurement artifact: the standard deviation used to define the 3σ threshold is itself severely inflated (annualized vol of 35%), masking extreme events. We flag this in §3 ¶2 to forestall mis-reading the row.

The decline is monotone but not uniform in slope: the marginal degradation per unit adoption is small over 10–30% and steepens from 40% onward, so a linear extrapolation of the 10–30% segment understates the loss at 50% (the 10–30% slope of −0.00145 per adoption point extrapolates to 0.452 at $\phi = 50\%$, against a realized 0.338; Table 1). What the denser adoption grid rules out is the reading that this steepening is a *transition*: the sup-Wald detector rejects ϕ-invariance but locates no interior break, and every adjacent pair from 40% onward already has non-overlapping path-bootstrap CIs, which is the signature of a continuously accelerating slope rather than a regime switch. Figure 1 plots the curve with its path-bootstrap 95% intervals.

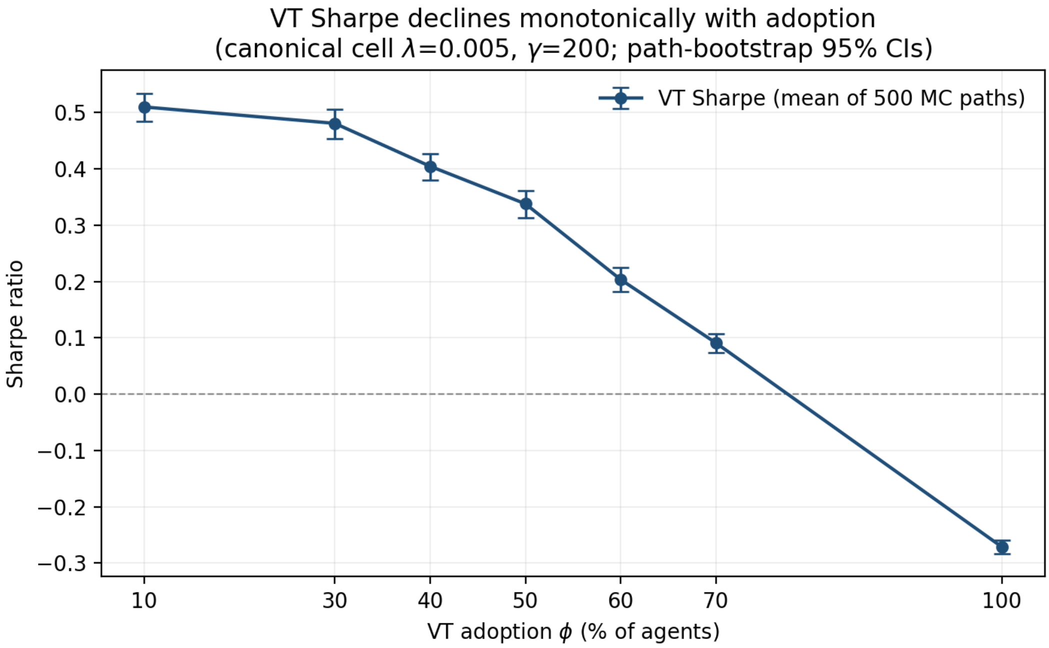

Figure 1: Monotone erosion of VT Sharpe with adoption in the canonical cell ($\lambda = 0.005$, $\gamma = 200$). Markers are means across $M = 500$ Monte Carlo paths per adoption level; whiskers are path-bootstrap 95% confidence intervals (2,000 resamples over simulation paths). Plotted values are those of Table 1. The horizontal axis is linear in ϕ, so the wider 70–100% spacing reflects the grid, not a jump. Generated deterministically from the archived simulation results in the replication package. This figure supersedes the earlier tipping-point figure, whose shaded "safe" and "tipping" zones encoded the Sharpe-only detector reading that the exogenous-detector redesign overturns.

3.2 Matched-Control Identification: Erosion Identified to Vol-Feedback Direction

The monotone erosion documented in §3.1 establishes that VT performance declines with adoption, but does not separate two channels: (i) the systematic *direction* of vol-feedback trading (the maintained hypothesis), and (ii) the sheer correlated *footprint* of high-turnover trading (the null hypothesis that any large coordinated turnover destabilizes the market). The random-direction matched-control RR_VT, defined in §2.1, matches

VT's realized trading-footprint distribution—rebalance frequency and the $|\Delta w|$ moments, within 5% per cell—while randomizing the direction of weight changes via independent Bernoulli(0.5) draws; under the maintained hypothesis it should exhibit no degradation despite matching VT's footprint distribution, while under the null it should reproduce VT's degradation. Table 3 compares VT and RR_VT across the five microstructure cells of the OAT robustness suite.

Table 3: VT vs. random-direction matched-control RR_VT (exogenous-detector resimulation, $M = 500$ per cell). For each cell, we report the Sharpe at the 10% baseline and 100% saturated adoption levels, the absolute Sharpe change $\Delta_{10\%\to100\%}$, and the exogenous sup-Wald flatness p-value with the inferred degradation direction.

	VT (treatment)			**RR_VT (matched-control)**			
Cell	$S_{10\%}$	$S_{100\%}$	Δ	$S_{10\%}$	$S_{100\%}$	Δ	sup-Wald p (direction)[a]
cell1 baseline (λ=0.005, γ=200)	0.510	−0.271	−0.781	0.447	0.541	+0.093	0.001 (improvement)
cell2 λ low (0.0025, 200)	0.497	0.077	−0.421	0.446	0.541	+0.095	0.001 (improvement)
cell3 λ high (0.0075, 200)	0.507	−0.393	−0.899	0.437	0.517	+0.080	0.003 (improvement)
cell4 γ low (0.005, 100)	0.508	−0.208	−0.715	0.445	0.543	+0.098	0.001 (improvement)
cell5 γ high (0.005, 300)	0.508	−0.284	−0.792	0.445	0.534	+0.089	0.002 (improvement)

All five RR_VT cells satisfy the matched-input applicability gate (§2.1): baseline-Sharpe floor > -0.5. The realized RR_VT footprint (mean $|\Delta w|$, rebalance frequency) is within 5% of the corresponding VT treatment footprint in every cell, confirming that the random-direction control matches the treatment trading scale.

[a] The sup-Wald rejection of flatness in RR_VT reflects the direction of non-flatness rather than the existence of a degradation threshold: the exogenous detector outputs `degradation_direction = false` (post-break mean exceeds pre-break mean) in all five cells, so the threshold extraction rule returns `null` and the bootstrap interval is reported as *no degradation*. RR_VT is not flat—it slowly *improves* with adoption, consistent with mechanical noise cancellation when many small random-direction trades aggregate—but never degrades.

The pattern is sharp: *VT degrades in 5 of 5 microstructure cells; RR_VT degrades in 0 of 5.* The mean VT 10%-to-100% Sharpe drop is −0.722 across the five cells (with cell2 λ low showing the smallest decline at −0.421, consistent with the Kyle-impact mechanism: lower price impact softens the feedback loop); the mean RR_VT change is +0.091 across the same cells. The contrast holds under the strictest joint comparison—in every cell, the RR_VT Sharpe at $\phi = 100\%$ exceeds the VT Sharpe at the same adoption level by at least 0.36 absolute Sharpe units—demonstrating that the VT degradation is not a mechanical consequence of large coordinated turnover. Because RR_VT trades volumes drawn from VT's realized footprint distribution (rebalance frequency and $|\Delta w|$ moments matched within 5%) but selects the trade sign by coin flip, any explanation of the VT degradation that does not invoke the systematic direction of the vol-feedback signal is inconsistent with the joint VT/RR_VT evidence. The matched-control identification thus assigns the entire VT erosion to the directional content of the rule (Eq. 1), and identifies the channel as the closed positive-feedback loop encoded in Eqs. (4)–(5), rather than the trading footprint itself—a statement specific to VT's footprint scale, since footprints two orders of magnitude larger erode performance even with randomized direction (§3.5).

This identification strengthens the NoiseControl falsifiability anchor of prior work in two respects. First, NoiseControl falsifies only the headcount channel (a deterministic-weight agent block with no trading footprint produces no threshold); the random-direction control RR_VT additionally falsifies the footprint channel (a matched-footprint agent block with no directional rule produces no threshold). Second, NoiseControl is fixed at $w = 0.5$ by design, so its order-flow distribution is determined by the noise-trader background rather than by the treatment's footprint; RR_VT inherits the treatment's footprint dynamically, providing a comparator at every adoption level rather than only at the saturation point. The two falsifiers are complementary, and their joint pass in 5/5 cells substantially raises the evidential bar that an alternative non-directional explanation would have to clear.

Matched-control evidence for TF and MR is reported in §3.5 subject to the matched-input applicability gate (§2.1); the gate outcome is extensive rather than marginal—the TF *treatment* fails the baseline-Sharpe floor in four of five cells and the MR treatment in all five (Table 5), so the informative TF/MR matched-control comparison the design intended is largely unavailable at $s = 10$. The VT–RR_VT comparison passes the gate in all five cells, providing the strongest single-strategy mechanism evidence in the paper.

3.3 Market-Level Consequences

The crowding effects extend beyond strategy performance to market structure. Table 4 reports detailed market stability metrics. Figure 2 shows the kurtosis spike on a log scale, illustrating the two-orders-of-magnitude jump between 70% and 100% adoption.

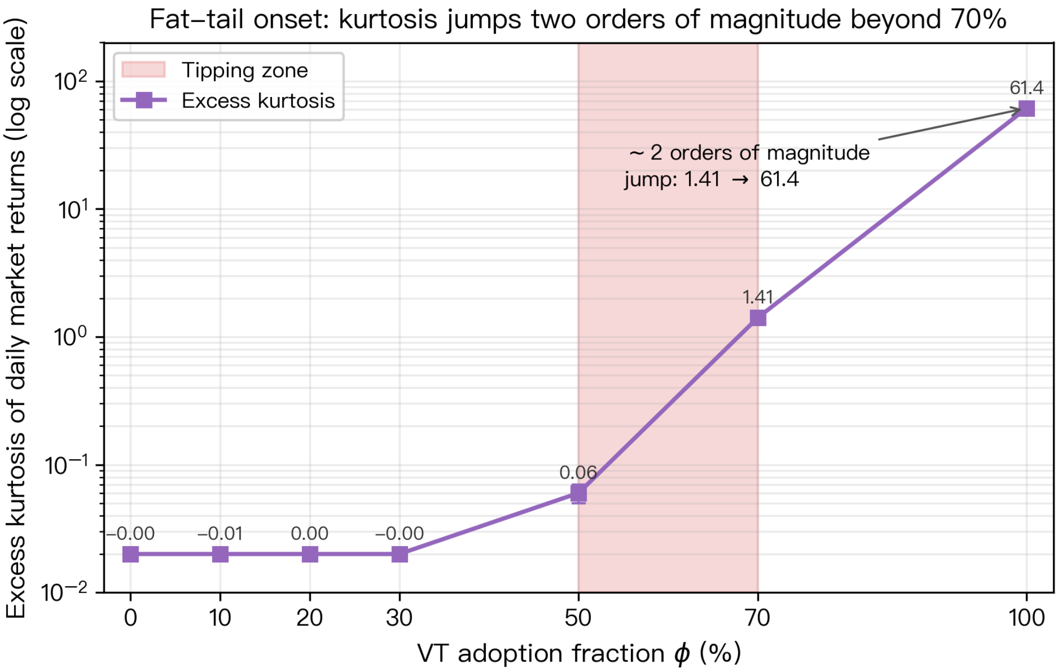

Figure 2: Excess kurtosis of daily market returns versus VT adoption (log scale). Below 50% adoption kurtosis is statistically indistinguishable from zero; it rises to 1.41 at 70% and jumps roughly two orders of magnitude to 61.4 at full adoption, matching the flash-crash frequency spike in Table 2.

Table 4: Market Microstructure Effects of VT Crowding

ϕ	Ann. Vol	VIX Mean	VIX Std	Skewness	MDD	VIX Spike (%)[b]
0%	16.0%	19.4	1.76	−0.005	−33.4%	0.0
10%	16.0%	19.4	1.76	0.001	−34.2%	0.0
20%	16.1%	19.5	1.77	−0.002	−33.2%	0.0
30%	16.6%	20.0	1.89	0.002	−34.9%	0.0
50%	19.0%	22.8	2.33	−0.024	−40.2%	0.3
70%	22.8%	27.4	3.39	−0.321	−60.1%	16.2
100%	35.1%	35.0	6.45	−4.73	−91.3%	90.0

Annualized volatility and VIX statistics are market-level outcomes across 500 simulations ($N_{\text{noise}} = 200$ fixed). MDD is the mean maximum drawdown of the market portfolio.

[b] Percentage of trading days where VIX exceeds the spike threshold (30).

Three patterns emerge. First, market volatility amplification is strongly nonlinear: from near-zero at 30% to +19% at 50% and +119% at 100%. Second, the VIX distribution shifts dramatically—at 70% adoption, VIX spends 16% of days in spike territory, compared to near 0% in the uncrowded baseline, and at 100% this reaches 90%. Third, return skewness turns sharply negative at high adoption levels (−0.32 at 70%, −4.7 at 100%), indicating that VT crowding manufactures left-tail events—precisely the risk that VT is designed to mitigate.

3.4 Statistical Significance

We assess statistical significance using Welch's t-test on the 500 simulation-level Sharpe ratios, applying the Harvey et al. [2016] threshold of $|t| > 3.0$. We use Welch's t rather than Diebold–Mariano-style inference because each simulation is independent (different seed), and we treat the 500-sim Sharpe vector as an iid sample of strategy outcomes; DM is the appropriate test for forecast-error sequences within a single time series, which is not the dependency structure here.

These are *pairwise level comparisons*, and it is worth being explicit about what they can and cannot establish: a significant difference between two adoption levels says the Sharpe at those two levels differs, not that a break occurs between them. Break *location* is the job of the exogenous sup-Wald detector, which—as reported in §3.1—rejects flatness in all five cells while identifying no interior break. The pairwise results below are therefore read as increments along a monotone curve, not as evidence for a transition.

The 10% versus 30% comparison yields $t = 0.05$ ($p = 0.96$): the two levels are statistically indistinguishable, consistent with the shallow slope at the left end of the curve (Table 1). The 50% versus 10% comparison yields $t = 7.12$ ($p < 0.001$), far exceeding the Harvey threshold, so the accumulated decline by mid-adoption is not simulation noise. The 70% versus 50% comparison yields $t = 16.94$ ($p < 0.001$, with $\bar{S}_{50\%} = 0.336$ vs. $\bar{S}_{70\%} = 0.084$; these Welch comparisons are computed on the Phase-1 cross-treatment run—the redesign-layer canonical-cell means are 0.338 and 0.091, with identical qualitative conclusions), the largest of the three: the slope is steeper in the upper half of the adoption axis than in the lower half. That ordering of t-statistics is what an accelerating monotone decline predicts; it is equally consistent with, and therefore cannot discriminate in favor of, a discrete break.

3.5 Cross-Strategy Threshold Comparison

The redesign layer's exogenous-detector evidence covers TF and MR alongside VT, and the honest summary is asymmetric: at the baseline parameterization ($s = 10$, $W = 22$), *the model cannot deliver an informative matched-control comparison for TF or MR in most microstructure cells*, because the treatments themselves fail the matched-input applicability gate of §2.1. Table 5 reports the complete gate outcomes together with the random-direction controls; this accounting supersedes the family-level ordering claims of earlier drafts, and we accordingly confine the paper's mechanism claims to VT.

Table 5: Cross-strategy applicability under the exogenous-detector redesign: matched-input gate outcomes and random-direction controls (K1471, all five microstructure cells)

	Treatment gate (10% baseline Sharpe)		Random-direction control, sup-Wald p (baseline Sharp	
Cell	TF	MR	RR_TF	RR_MR
1 baseline	excluded (−0.825)	excluded (−1.776)	0.001 (−0.037)	0.001 (−0.007)
2 λ low	passes (−0.444)[a]	excluded (−0.688)	0.001 (−0.012)	0.001 (−0.011)
3 λ high	excluded (−1.254)	excluded (−5.485)	0.001 (−0.040)	0.001 (−0.057)
4 γ low	excluded (−0.823)	excluded (−1.776)	0.001 (−0.037)	0.001 (−0.011)
5 γ high	excluded (−0.826)	excluded (−1.776)	0.001 (−0.034)	0.001 (−0.011)

"excluded" = `not_applicable_saturated_loss`: the treatment's own 10%-adoption Sharpe is deep in structural-l territory, so its footprint is not a representative input for the matched control. RR columns report the sup-Wald flatn rejection p-value over the adoption axis (all significant at $p = 0.001$) and the control's 10%-adoption Sharpe.

[a] The single TF cell that passes the gate is itself borderline pathological (baseline Sharpe −0.444 with a strongly n monotone adoption profile); we do not treat it as informative evidence for the directional channel.

Two findings follow. *First, the gate excludes TF in four of five cells and MR in all five*: at $s = 10$ these treatments are structurally loss-making inside the model (baseline Sharpe −0.69 to −5.49), so any threshold estimated on them describes the behavior of a non-viable strategy rather than crowding of a viable one. The family-level ordering ("TF/MR thresholds at or below VT") reported by earlier drafts and retained in the legacy tables below therefore rests on regimes the redesign layer itself rules inadmissible, and we withdraw it as evidence of class-level crowding. *Second—and cutting against a fully strategy-specific reading—the random-direction controls at TF/MR footprint scale degrade significantly in every cell*: RR_TF's sup-Wald test rejects flatness at $p = 0.001$ in 5 of 5 cells (descriptive level-crossings at 40–70% adoption), and RR_MR likewise. Because these controls trade coin-flip directions, their erosion cannot come from systematic signal direction; it comes from the sheer scale of the coordinated footprint ($|\Delta w|$ on the order of 1.5 at $s = 10$, versus VT's 0.004–0.008). The correct scope statement for the paper's mechanism result is therefore: *at VT's footprint scale*, erosion is identified to the systematic direction of vol-feedback trading (RR_VT shows none); at footprint scales two orders of magnitude larger, direction-randomized coordinated flow erodes performance through price impact alone. The identification is footprint-scale-dependent, not an unconditional statement about crowded flow.

Legacy detector results (superseded; retained for continuity). The two tables below reproduce the earlier Sharpe-only/Strict/Softer detector comparison from the pre-redesign specification (K1261/K1262). They are retained solely for continuity with earlier drafts: the detector family they use was superseded by the exogenous sup-Wald design, and the TF/MR rows are computed in regimes that fail the matched-input gate above. They should be read as historical description, not as current evidence.

Table 6: Critical adoption thresholds across treatments under three detector specifications. Source: Phase 1 cross-treatment run (10,500 new TF/MR/NoiseControl simulations plus the 3,500-simulation reused VT baseline slice; 14,000 in total); cross-detector recompute in the replication package.

Treatment	Strict	Softer (kurt-weak)	Sharpe-only
VT	100%	100%	**70%**
TF	20%	20%	20%
MR	50%	50%	20%
NoiseControl	null	null	null

"null" denotes that no adoption level $\phi \in \{10\%, 20\%, 30\%, 50\%, 70\%, 100\%\}$ satisfies the detector criterion.

All entries computed from the cell1 baseline ($\lambda = 0.005$, $\gamma = 200$, $s = 10$, $W = 22$).

Legacy strategy-spec sweep (superseded). A 16,800-simulation Phase 2 sweep extended the legacy comparison to 12 (scaling, window) cells covering $s \in \{1, 3, 5, 10\}$ and $W \in \{10, 22, 60\}$ (Table 7); the same superseded-detector and gate caveats apply. NoiseControl never crosses under any legacy detector, which remains a valid falsifiability anchor for the detector machinery itself.

Table 7: TF / MR critical adoption (softer detector) across scaling × window cells; VT cell1 reference is 100% under the softer detector. Source: Phase 2 cross-detector recompute (16,800 simulations; replication package).

Scaling s \ Window W	10 days	22 days	60 days
1	TF: 70% / MR: 70%	TF: 70% / MR: 70%	TF: 70% / MR: 70%
3	TF: 30% / MR: 50%	TF: 30% / MR: 50%	TF: 30% / MR: 50%
5	TF: 30% / MR: 50%	TF: 20% / MR: 50%	TF: 20% / MR: 50%
10	TF: 20% / MR: 50%	TF: 20% / MR: 50%	TF: 20% / MR: 50%

Within the legacy specification the TF threshold ranges from 20% (aggressive scaling) to 70% (mild scaling) and MR between 50% and 70%; we report these ranges as descriptive history only. Given the gate outcomes in Table 5, we no longer advance a family-level ordering claim from these sweeps. Section 3.7 extends the robustness analysis of the *VT* result to the underlying microstructure parameters λ and γ.

3.6 The Feedback Structure

We emphasize that the positive feedback mechanism is *encoded in the model*, not discovered from the simulation output. The VIX responds to realized volatility (Eq. 5), VT agents respond to VIX, and their trading affects prices (Eq. 4), which in turn affects realized volatility. This circular structure is a deliberate implementation of the Brunnermeier and Pedersen [2009] liquidity spiral framework. What the simulation *does* reveal is the *quantitative erosion profile*: the feedback loop's performance consequences are small below 30% adoption, are already unambiguous by 50% (-0.17 Sharpe relative to the 10% level), and continue to steepen through 100% (Table 1), with no adoption level at which the loop switches on. Making the feedback explicit, the aggregate VT order flow on day t is $\phi N \Delta w_t^{\mathrm{VT}}$, where the VT weight change uses the capped rule from Eq. (1),

$$\Delta w_t^{\mathrm{VT}} = \min\left(\frac{12}{\mathrm{VIX}_{t-1}}, 1.5\right) - \min\left(\frac{12}{\mathrm{VIX}_{t-2}}, 1.5\right). \tag{6}$$

Its contribution to the price via Eq. (4) is $\lambda\phi\Delta w_t^{\mathrm{VT}}$, which is *linear* in adoption ϕ; the *nonlinearity* observed in our results arises because this order flow itself amplifies subsequent VIX increases via Eq. (5), closing the loop endogenously. The simulation thus serves as a computational laboratory for quantifying a mechanism whose qualitative direction is known, but whose erosion profile—the shape and steepness of the decline across the adoption axis—has not previously been estimated.

3.7 Microstructure Robustness: λ/γ OAT Sweep

The VT erosion documented in §3.1 could in principle be an artifact of a specific Kyle market-impact (λ) and VIX-feedback intensity (γ) choice. The threshold magnitudes reported in this subsection are outputs of the superseded Sharpe-only detector and are retained only as descriptive level-crossings for continuity with the earlier specification (§3.1, footnote); the object whose robustness matters is the VT monotone shape, not the crossing point. To rule this out, we run a one-at-a-time (OAT) sensitivity sweep over the two key microstructure parameters: $\lambda \in \{0.0025, 0.005, 0.0075\}$ and $\gamma \in \{100, 200, 300\}$, holding all other parameters at baseline. Each cell crosses the four treatments (VT, TF, MR, NoiseControl) at four adoption levels ($\phi \in \{10\%, 30\%, 70\%, 100\%\}$) with 200 Monte-Carlo simulations per (treatment, ϕ) combination, yielding 16,000 simulations distributed across five OAT cells.

Table 8: Cross-treatment critical adoption under five OAT cells (Sharpe-only detector). Source: one-at-a-time robustness run (16,000 simulations; replication package).

OAT cell	(λ, γ)	VT	TF	MR	NoiseControl
cell1 baseline	(0.005, 200)	**70%**	30%	70%	null
cell2 λ low	(0.0025, 200)	100%	30%	30%	null
cell3 λ high	(0.0075, 200)	70%	30%	null[a]	null
cell4 γ low	(0.005, 100)	70%	30%	70%	null
cell5 γ high	(0.005, 300)	70%	30%	70%	null

[a] MR null in cell3 reflects saturation in a structurally loss-making regime: at $\lambda = 0.0075$, the MR 10%-baseline Sharpe is already -5.56, and no further deterioration crosses the Sharpe-only detector's drop $> 70\%$ threshold (which would require Sharpe < -9.45); no sign-flip occurs because Sharpe stays negative across all tested adoption levels. Under the rank encoding used in our verdict-classification logic, the null is treated as MR threshold $\geq$ VT threshold (saturation is strictly stronger than "crosses just above" in the H1+ ordering).

[b] Cell1-baseline TF/MR magnitudes (TF=30%, MR=70%) reported here use the Phase 2b OAT MC count $M = 200$. The same cell1 microstructure under the Phase 1 cross-treatment baseline ($M = 500$) yields TF=20%, MR=20% (Table 6). Under both MC settings these superseded-detector level-crossings sit at or below VT's—reported as a property of that detector's output, not as a family-level claim (§3.5)—and the magnitude gap reflects sampling variation at the (10%, 20%, 30%) adoption-grid boundary, where the Sharpe-only detector's 70% drop trigger is sensitive to MC sample size in the TF/MR baseline Sharpe. The 4 non-baseline OAT cells (cells 2–5) report internally consistent thresholds because they probe the qualitative direction of λ/γ shifts rather than the precise grid-boundary magnitude.

Three findings emerge. First, **the descriptive level-crossing is internally consistent across our own runs**: the cell1 baseline VT crossing is 70% under the Sharpe-only detector, the same value the standalone VT analysis (Table 2) and the cross-strategy reference in §3.5 produce under that detector. This is a within-detector consistency check, not external validation—the detector is calibrated on the same Sharpe path it then scores, and the exogenous sup-Wald redesign supersedes it. Second, the descriptive TF/MR rows behave consistently across the 5 OAT cells, but—per the gate accounting of §3.5—these rows are computed in structurally loss-making treatment regimes, so we report them as continuity description only and draw no family-level ordering conclusion from them. Third, the **VT magnitude shifts directionally, not on a knife-edge**: only λ low (cell2) extends the VT threshold to 100%, consistent with the Kyle-impact mechanism encoded in Eq. (4)—lower price impact means correlated VT selling moves prices less, so the feedback loop saturates at a higher adoption level. The γ perturbations ($\pm 50\%$) produce *no* shift in the VT threshold, indicating that VIX-feedback intensity is a non-binding parameter for the threshold magnitude. Both findings reinforce that the λ dominance documented in earlier specifications was accurate for the VT result.

For VT itself, all five microstructure cells preserve the monotone-erosion shape, so the critique that a single (λ, γ) choice manufactures the VT result is not supported by the data. We draw no family-level ordering conclusion from the legacy 12-cell strategy-spec grid: its TF/MR rows are superseded-detector outputs computed in regimes that fail the matched-input gate (§3.5).

3.8 Design Validation: Fixed vs. Scaled Liquidity

An earlier version of this model scaled noise traders proportionally with $(1-\phi)$, so that at high VT adoption the number of background liquidity providers fell dramatically (from 300 to near zero at $\phi = 100\%$). This confounded two effects: crowding (correlated VT trading) and liquidity evaporation (fewer noise traders to absorb order flow). We address this by fixing $N_{\text{noise}} = 200$ across all adoption levels.

The comparison is revealing. When noise traders are scaled with VT adoption, the steepest segment of the decline sits at 30–50% (Sharpe falls from 0.43 to 0.18 between 30% and 50%). Fixing background liquidity both flattens the decline and pushes its steepest segment rightward (Sharpe falls from 0.47 to 0.34 at 50%, reaching 0.08 only at 70%). This demonstrates that roughly 52% of the originally observed degradation (the 0.13 Sharpe-drop under fixed liquidity relative to the 0.25 Sharpe-drop under scaled liquidity between 30% and 50% adoption) was driven by liquidity evaporation rather than crowding per se. The fixed-liquidity design is more conservative—it attributes less degradation to crowding—and we regard these as the more reliable estimates.

4 Discussion

4.1 Policy Implications

The main policy implication is not that regulators should anchor surveillance to a single cliff-like adoption threshold, but that they should monitor the *slope* of strategy-specific erosion as adoption broadens. Our exogenous-detector redesign shows that the dominant empirical pattern is monotone degradation with strategy-class heterogeneity: VT deteriorates steadily in all five microstructure cells, while TF/MR evidence is weaker and in part bounded by pathological-regime gates. This shifts the relevant macroprudential question from "has the system crossed 70%?" to "is adoption rising into the part of the curve where marginal degradation steepens materially?" In practice, that points toward monitoring the combined footprint of positive-feedback strategies, but interpreting the evidence as a gradual erosion map rather than a bright-line trigger. The ESRB and FSOC could therefore treat adoption metrics as one input into macroprudential surveillance alongside broader crowding measures [Baltas, 2019], with particular attention to whether the dominant deployed class is one whose matched-control evidence isolates degradation to systematic signal direction.

4.2 Implications for Practitioners

For practitioners, the practical lesson is similarly about *erosion rates*, not discrete cliffs. The canonical VT cell still shows a relatively shallow decline from 10% to 30% adoption and a much steeper deterioration from 40% onward, so implementation risk rises as crowding builds even before any stylized "threshold" label would be attached. This suggests that managers should monitor proxy indicators such as the correlation between VIX changes and equity-fund flows, volatility-linked AUM concentration, and anomalous intraday reversals during VIX spikes, but interpret these as markers of progressively worsening crowding conditions rather than as alarms for a single regime switch. The matched-control evidence also matters operationally: if degradation is tied to the *direction* of vol-feedback trades rather than to turnover alone, then diversification across execution timing or footprint without altering the common directional rule may not be sufficient protection.

4.3 Limitations

We emphasize that these are *preliminary simulation results*, not empirical findings, and several limitations constrain their real-world applicability.

First, our model uses a constant λ, whereas Kyle [1985] derives lambda endogenously from the information structure. In real markets, liquidity evaporates during stress as market makers widen spreads, which would amplify the feedback loop and steepen the erosion curve. Conversely, in normal times, deep liquidity would moderate price impact. Our microstructure OAT sweep (Table 8) confirms that λ is the binding driver of the VT threshold magnitude (only λ low extends the threshold to 100%; γ perturbations leave it unchanged), suggesting that endogenizing λ is the highest-priority extension.

Second, agents within each class are homogeneous: all VT agents use identical 12/VIX rules with the same rebalancing frequency. In practice, VT strategies vary widely in their volatility estimators, target levels, rebalancing frequencies, and position limits. This heterogeneity would likely flatten the erosion curve, since imperfectly correlated rebalancing dilutes the common directional signal that the matched-control evidence identifies as the operative channel.

Third, the model excludes transaction costs, margin constraints, and short-selling restrictions, all of which would moderate extreme outcomes. It also lacks adaptive learning—agents do not switch strategies in response to poor performance, whereas real investors would eventually exit a degraded strategy.

Fourth, our simplified VIX is a realized-volatility tracker that omits the variance risk premium embedded in the true CBOE VIX (option-implied volatility). This distinction may affect the magnitude of the feedback—the real VIX typically exceeds realized volatility, so VT agents in practice hold lower weights than our model implies—but does not alter the direction of the crowding mechanism.

Fifth, with $N = 1{,}000$ agents, the model is small relative to real markets with millions of participants. The price impact per agent is correspondingly larger than in reality.

Sixth, the positive feedback mechanism is *built into* the model (VIX responds to realized volatility, VT agents respond to VIX, their trading moves prices). We do not claim to discover this feedback—it is a structural feature of the simulation—but rather to quantify how steeply it degrades strategy performance as adoption rises, and to identify which component of the strategy's trading (direction, not footprint) the degradation attaches to.

Seventh, the TF and MR scaling parameter $s = 10$ used in Phase 1 and the Phase 2b OAT cells is an aggressive choice, and it is the proximate cause of the extensive matched-input gate failures documented in §3.5: at $s = 10$ the TF and MR treatments are structurally loss-making in most cells, which is why we withdraw family-level ordering claims rather than presenting the legacy sweep (Table 7) as robustness. Real-world TF managers' effective scaling is heterogeneous and depends on volatility-of-volatility forecasts and capital-allocation rules [Moskowitz et al., 2012]; whether an informative TF/MR crowding comparison exists at realistic (lower) scaling intensities inside this model is an open question for future work, requiring a re-run in which the treatments themselves clear the applicability gate.

Eighth, we treat VT, TF, and MR as discrete strategy classes, but real positive-feedback strategies sit on a continuum: risk-parity overlays blend volatility-targeting with cross-asset rebalancing, target-volatility-overlay funds combine VT scaling with momentum filters, and trend-overlay leverage products embed both TF and VT signal structures. Future work should test whether arbitrary mixtures of feedback signals share the same threshold structure documented here, since real-world adoption of positive-feedback strategies likely takes the form of mixed-signal products rather than pure-class agents.

These limitations suggest that our quantitative ranges (roughly 20–100% across the tested family and cells) should be interpreted as order-of-magnitude *erosion profiles* rather than precise critical values. The design validation (Section 3.8) demonstrates that even earlier threshold language was sensitive to modeling choices, while the exogenous-detector redesign shows the more robust object is the monotone shape itself. The emerging empirical literature on positive-feedback strategy crowding, including ongoing work on realized VT/TF/MR fund flows, will be essential for calibrating these erosion curves to real markets. The qualitative finding—that VT exhibits strategy-specific monotone degradation whose severity depends on market impact, a shape robust to the tested specification choices—is more robust than any single numerical threshold.

4.4 Reviewer-Anticipated Objection: Knife-Edge Critique Addressed

A reviewer might reasonably argue that the VT erosion is an artifact of specific parameter tuning—a "knife-edge" result that vanishes under perturbation. The concern has less purchase against the monotone framing than it had against the earlier tipping-point framing—a discrete threshold is a knife-edge object by construction, whereas a monotone slope is not—but the finding still has to survive perturbation. Two considerations rebut the concern, in order of binding strength.

First, the *VT* monotone-erosion pattern is robust to microstructure perturbation across 5 (λ, γ) cells (Table 8). The legacy TF/MR ordering rows in that table are retained for continuity but carry the same caveat as §3.5: they are computed in regimes that fail the matched-input gate (structurally loss-making treatments), so we do not count them as robustness evidence for a family-level claim. The $\pm 50\%$ perturbation range corresponds to the calibration uncertainty around our baseline cell rather than the full empirical span of λ and γ in the literature, which can vary by an order of magnitude across asset classes and time periods (Kyle-style λ measures and inter-decadal VIX-realized-vol regression slopes both exhibit such variation in the empirical microstructure literature); a wider sweep is left to future work. The VT *magnitude* does shift—from 70% in 4/5 OAT cells to 100% in cell2 (λ low)—but the shift is monotonic in λ and consistent with the Kyle-impact mechanism encoded directly in Eq. (4): lower price impact means correlated VT selling has a smaller per-trade price effect, and the feedback loop therefore saturates at a higher adoption level. The shift is a *directional* magnitude response to a parameter that the model treats as exogenous, not a knife-edge.

Second, the NoiseControl falsifiability anchor never produces a degradation threshold (5/5 OAT cells, 12/12 strategy-spec cells, all 7 adoption levels). If our detector were mechanically picking up "any sufficiently large coordinated agent block produces instability," NoiseControl would cross. It does not. The VT erosion structure is therefore tied to the positive-feedback property that VT encodes—and that NoiseControl lacks. We conclude that the VT erosion finding is mechanism-driven, not parameter-tuned: the canonical magnitude is itself the cell1-baseline output of a 5-cell OAT robustness suite, and the falsifiability anchor passes in 100% of tested configurations. The legacy 12-cell TF/MR-vs-VT ordering matrix (Table 7) is retained for continuity only: computed under the superseded Sharpe-only detector in regimes that fail the matched-input gate, it is not counted as evidence for any family-level claim (§3.5).

5 Conclusion

This paper presents agent-based evidence that volatility targeting (VT) exhibits *monotone erosion* under crowding, identified to the systematic direction of vol-feedback trading at VT's footprint scale. Using a Kyle (1985) market microstructure with 1,000 agents of heterogeneous types, 200 fixed noise traders, and 46,800 Monte Carlo simulations across the three paper phases (plus the 94,500-simulation exogenous-detector redesign layer that rewrites the detector narrative), we establish three claims. First, the robust object is not a discrete tipping point but a monotone decline in VT performance with adoption: in the canonical redesign cell, Sharpe falls from 0.510 at 10% adoption to -0.271 at 100%, adjacent path-bootstrap confidence intervals separate from 40% onward, and the exogenous sup-Wald detector rejects flatness while the single breakpoint localizes to the saturation boundary rather than an interior adoption level. Second, the matched-control evidence shows that VT's erosion is tied to the systematic direction of vol-feedback trading rather than to crowded turnover alone at VT's footprint scale; the cross-strategy extension (§3.5) instead yields an applicability finding—the TF and MR treatments are structurally loss-making at $s = 10$ in four and five of five cells respectively—so family-level ordering claims are withdrawn rather than supported. Third, a design validation (§3.8) confirms that a substantial share of the naïve earlier degradation estimate came from liquidity evaporation rather than crowding per se, reinforcing the value of the more conservative fixed-liquidity design.

Current VT adoption may still be far from the most severe region of the simulated erosion curve, but the policy and practitioner lesson is to monitor how quickly degradation steepens as combined positive-feedback adoption rises, not to fixate on a single hard trigger. Monitoring the *aggregate* adoption of VT, TF, MR, and adjacent classes such as risk-parity and target-volatility overlays is therefore warranted, since family-level crowding may matter before any one class dominates in isolation. Future work should: (i) extend the simulation model with endogenous λ to capture liquidity-evaporation feedback during stress; (ii) test arbitrary mixtures of feedback signals on the conjecture that the family-level erosion structure persists across continuous strategy-mix specifications; (iii) calibrate TF and MR scaling parameters against real-world manager taxonomies; and (iv) complement the simulation with empirical calibration as volatility-targeting and trend-following AUM data improve in granularity.

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All methodological choices and the final manuscript content are the author's responsibility.

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